Algebra: Chapter 0 Solutions

Mango's Reading Group

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Chapter I

Preliminaries: Set theory and categories

1 Naive Set theory

Ex 1.1. Russell's Paradox shows that naïve set theory's assumption of the unrestricted comprehension principle – for any property P, the set $\{x : P(x)\}$ exists – leads to a contradiction. Consider the property $P(x) : x \in x$. The unrestricted comprehension prinicple tells us that the set $S = \{x :\sim P(x)\}$ exists. This is simply not possible, since we get $S \in S \iff S \notin S$.
To see this, note that if S contains itself, then S does not satisfy $\sim P(S)$, so $S \notin S$, but now S does satisfy $\sim P(S)$, so $S \in S$. i.e, $S \in S \iff S \notin S$, and we know that only one of those alternatives is possible. A more detailed discussion can be found at https://plato.stanford.edu/archives/fall2025/entries/russell-paradox/

Ex 1.2. First, we check that the elements of $\mathscr{P}_\sim$ are nonempty. Indeed, since $\sim$ is an equivalence relation, we have that for all $a \in S : a \sim a$, so that $a \in [a]$ i.e $[a]$ is nonempty. Now we check that distinct elements of $\mathscr{P}_\sim$ are disjoint. Indeed, let $[a] \neq [b]$ be two elements of $\mathscr{P}_\sim$. If they did intersect at some c, we would have $a \sim c$ and $b \sim c$ which by transitivity would give $a \sim b$. This is simply not possible, because this implies by transitivity that for every $x \in S$ such that $x \sim b$, we have $x \sim a$. i.e $[b] \subseteq [a]$. Symmetry gives $[a] \subseteq [b]$ i.e $[a] = [b]$, which contradicts our initial assumption of $[a] \neq [b]$. Ofcourse the union of the elements of $\mathscr{P}_\sim$ is S because the elements of $\mathscr{P}_\sim$ are $[a] \supseteq \{a\}$ for each $a \in S$. This gives us

$$\bigcup_{a \in S} [a] \supseteq S.$$

The other inclusion is clear, since each $[a]$ is a subset of S. Thus we have $\bigcup_{x \in \mathscr{P}_\sim} x = \bigcup_{a \in S} [a] = S$ as desired. i.e, $\mathscr{P}_\sim$ is a partition of S.

Ex 1.3. let $\mathscr{P} = \{S_\alpha : \alpha \in I\}$ be a partition of S. We claim that the equivalence relation $\sim$ given by $a \sim b \iff (\exists \alpha \in I : a \in S_\alpha \text{ and } b \in S_\alpha)$. I.e two elements are related iff they lie in the same part of the partition. Indeed, this is an equivalence relation. For any $a \in S$, we have $a \sim a$ because $a \in S_\alpha$ for some α so $\sim$ is reflexive. Now suppose that $a \sim b$ and $b \sim c$. Since b lies in exactly one S_α, we have that a and c lie in the same S_α, i.e $a \sim c$ so $\sim$ is transitive. Symmetry is clear, if a lies in the same part as b, b lies in the same part as a.

Ex 1.4. From what we have seen, it suffices to count the number of partitions of $\{1, 2, 3\}$, which are $\{\{1\}, \{2\}, \{3\}\}, \{\{1, 2\}, \{3\}\}, \{\{1, 3\}, \{2\}\}, \{\{2, 3\}, \{1\}\}, \{\{1, 2, 3\}\}$. I.e, there are 5 equivalence relations on the set $\{1, 2, 3\}$

Ex 1.5. On the set $\{1, 2, 3\}$ define the following equivalence relation:

$$1 \sim 1, 2 \sim 2, 3 \sim 3, 3 \sim 2, 2 \sim 1, 2 \sim 3, 1 \sim 2.$$

This is reflexive and symmetric but not transitive. Indeed, $1 \sim 2$ and $2 \sim 3$ but $1 \not\sim 3$. If we use a relation that is only reflexive and symmetric but not transitive, then we will not get a partition. Indeed, the parts may intersect. For example, in the above relation we defined, if we let $[a] := \{b : a \sim b\}$, then $[1] \cap [2] = \{2\}$ even though $[1] \neq [2]$.

Ex 1.6. The relation $\sim$ is reflexive since indeed, $a - a = 0 \in \mathbb{Z}$ for all $a \in \mathbb{R}$. It is symmetric, since the negative of an integer is an integer. Indeed, $a \sim b \implies a - b = k_1 \implies b - a = -k_1$. It is transitive, since the sum of two integers is an integer. Indeed, if $a \sim b$, $b \sim c$, we have $a - b = k_1, b - c = k_2$ for some $k_1, k_2 \in \mathbb{Z}$ which gives $a - c = k_1 + k_2 \in \mathbb{Z}$. Hence, it is an equivalence relation. In exactly the same way, we get that the relation $(a_1, a_2) \approx (b_1, b_2) \iff a_1 - b_1 \in \mathbb{Z}$ and $a_2 - b_2 \in \mathbb{Z}$ is an equivalence relation.
As for finding “compelling” descriptions, first relation can be seen geometrically as a ‘circle’ in the sense that the set $\mathbb{R}/\sim$ can be identitfied with the set $\{e^{2\pi i x}\}$ via $[x] \mapsto e^{2\pi i x}$. It is important to note that since we are using the representative of the classes in defining our correspondence, that it is well defined. Similarly, the set $\mathbb{R} \times \mathbb{R}/\approx$ can be identitifed with a torus, via the same exponential map: $[(x, y)] \mapsto (e^{2\pi i x}, e^{2\pi i y})$ which one can again check is well defined.